

TrueTone

An AI skin-scan that works on every face.

Competitive teardown · MVP scope · business model · the regulatory landmines
Prepared as a co-founder working document — to align us, then brief the developer.

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Working codename: TrueTone (final name TBD)
Date: June 2026

READ SECTION 9 BEFORE WRITING ANY CODE
(Biometric law. You're in Illinois. It matters.)

SECTION 01

The opportunity, in one page

Here's the thesis in plain terms. People buy skincare almost blind — they pick a serum off a shelf or a model's face, hope it suits them, and return it when it doesn't. AI face-scanning fixes that by turning a selfie into a read on the skin and a specific routine. The technology now works well enough on a phone, the market is large and growing fast, and — this is the part that matters — **the incumbents all share the same blind spot.**

Independent research is consistent and damning: skin-analysis models trained on the standard datasets perform measurably worse on darker skin tones (Fitzpatrick IV–VI), because those datasets are skewed heavily toward light skin. There is also a “forgotten middle” (Fitzpatrick III–IV — much of the East Asian, Hispanic, South Asian and Mediterranean population) that is neither the focus of fairness work nor well represented. Nearly every competitor markets “works on all skin tones”; the literature says most of them are overstating it. That gap between the marketing claim and the measured reality **is our wedge.**

THE ONE-SENTENCE PITCH

“The skincare AI that actually works on your skin — validated to perform equally across every skin tone, and honest about what it can and can't see.”

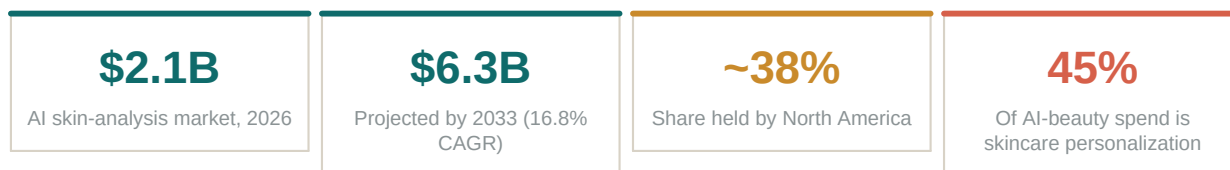
Why this is winnable by a small team

- **The moat is data composition, not model size.** We don't need to out-engineer L'Oréal. We need a deliberately balanced, consented dataset and per-skin-tone validation — something the incumbents are institutionally bad at because they bolted skin tone on late.
- **Trust is unoccupied.** Most of these tools are conversion engines that exist to sell a brand's products. A genuinely brand-neutral recommender is a real, defensible position (only one competitor, Lovi, is seriously playing here).
- **The build is cheap to start.** A credible v0 rides on existing vision models + a strong recommendation layer. We are not training a foundation model on day one (see Section 8).
- **Regulation is a moat if you treat it as one.** The same biometric-privacy rules that scare competitors into vague compliance become a trust feature when we lead with them (Section 9).

SECTION 02

Market size — the numbers for the deck

Three nested markets matter. The narrow one (AI skin analysis specifically) is the cleanest comparison for us; the wider ones show the pull of the category. Treat analyst figures as directional, not gospel — they vary by methodology — but the direction is unambiguous: up and to the right, led by North America.



The surrounding categories (context, not our TAM)

- **Beauty tech overall:** ~\$66B (2024) heading toward ~\$173B by 2030; AI is the fastest-growing slice and already ~a third of it.
- **Personalized skincare products:** ~\$33B (2025) — this is the spend our recommendations ultimately influence, and the basis for any commerce revenue.
- **The honest read:** we are not chasing a \$66B number. We're chasing the AI-skin-analysis slice plus a share of the personalized-product spend our routine drives. That's the figure to model.

CO-FOUNDER NOTE ON THE NUMBERS

Every market-research firm publishes a different total; I've used the more conservative, recently dated ones. When we fundraise, cite the AI-skin-analysis CAGR (it's defensible) and avoid the \$30B+ headline TAMs — sophisticated investors discount those instantly.

SECTION 03

Competitive landscape

The field looks crowded, but it sorts cleanly into three tiers that compete for different buyers. Knowing which tier we're in keeps us from fighting the wrong war.

Tier 1 — B2B infrastructure (“picks & shovels”)

Players: Haut.AI, Perfect Corp / YouCam, Revieve, GlamAR, IQONIC.AI, L'Oréal ModiFace & SkinConsult, Vichy.

They sell SDKs/APIs that power skin-analysis on brand websites and in-store kiosks. Perfect Corp alone claims 600+ brand customers; Haut.AI advertises 3M+ training data points and 20+ metrics. Deep-pocketed, enterprise-focused, and largely invisible to consumers. We do NOT want to fight them head-on as an infrastructure vendor on day one — but they are a plausible acquirer and a future B2B2C channel for us.

Tier 2 — Consumer apps (our actual arena)

Players: Lovi, Freya, Glo/Glow, Glass, Skin Bliss, TroveSkin, OnSkin, Derm AI, SkinCircle, Cureskin.

Direct-to-user apps that scan a selfie, score a handful of concerns (hydration, texture, acne, dark spots, aging), generate a routine, and track progress. Differentiation is mostly tone: Lovi = brand-neutral / anti-hype; Freya = sensitivity focus; Glo/Glass = radical simplicity. Most monetize via subscription and/or product affiliate. **This is where we compete.**

Tier 3 — Clinical / medical-grade (regulated)

Players: Skin Analytics, DermaSensor, FotoFinder, First Derm, MetaOptima / DermEngine, Canfield.

These detect or triage disease (notably skin cancer) and are regulated medical devices with clearances. High barrier, slow, expensive. **We deliberately stay out of this tier** — crossing into diagnosis is what turns a fun app into an FDA submission (Section 9).

What the best consumer apps already do well

We should assume parity on these and not pretend they're innovations: live-camera or photo scan; 8–20 concern scores; auto-generated AM/PM routines; progress photos over time; ingredient lookups; a conversational “skin assistant” chat. If we ship without these we look behind, not ahead. Our edge has to come from the *quality and honesty* of the read — not the feature checklist.

SECTION 04

Where everyone is weak — our five openings

These are the cracks the research and the user reviews keep exposing. Each is a feature we lead with, not a footnote.

1. SKIN-TONE EQUITY (THE BIG ONE)

The gap: Models underperform on Fitzpatrick IV–VI and on the intermediate III–IV “forgotten middle.” Standard datasets are 70%+ light skin. Competitors claim universality they can’t substantiate.

Our move: Build a deliberately balanced, consented dataset; report accuracy *per skin-tone band*; make “validated across all six Fitzpatrick types” a headline, with the receipts. This is the brand.

2. TRUST & NEUTRALITY

The gap: Most apps are storefronts — recommendations follow affiliate margin, not skin need. Users have caught on and resent it.

Our move: Brand-neutral scoring by default. Recommend by ingredient/formulation fit first; show *why* a product was picked; clearly label anything sponsored. Trust is the retention engine.

3. HONEST PROGRESS TRACKING

The gap: Lighting, angle, and camera differences make most “your skin improved 23%” claims noise. Users stop believing the graph, then churn.

Our move: Standardized capture (guided framing, lighting check, reference card option) so week-over-week comparisons are real. Under-promise, then actually deliver a trustworthy trend.

4. PRIVACY AS A FEATURE

The gap: Face data = biometric liability. Industry analysts flag BIPA/GDPR compliance — not accuracy — as the real barrier to scaling. Most apps bury this.

Our move: On-device or ephemeral processing where feasible; “we never sell your face” as a promise; one-tap delete. Lead with it (Section 9). It de-risks us AND differentiates.

5. CLOSING THE LOOP ON OUTCOMES

The gap: Apps hand you a routine and a vanity score, but rarely prove the routine worked for someone like you. No feedback loop = no compounding advantage.

Our move: Capture outcomes (“did this help?” + follow-up scans) to learn what works per skin-type cohort. This is the flywheel that makes our recommendations get smarter than theirs over time.

SECTION 05

Our wedge & positioning

Positioning is a series of refusals as much as promises. Here's who we are.

WE ARE	WE ARE NOT
A consumer skin-coach that gives an honest read and a brand-neutral routine.	A storefront that nudges you toward whatever pays us the most.
Validated to work across every skin tone — and transparent about it.	Another “works for all” claim with no evidence behind it.
A general-wellness / cosmetic tool: appearance, hydration, texture, routine.	A diagnostic medical device. We never say “you have [disease].”
Privacy-first: your face data is yours; deletable; never sold.	A data-broker dressed up as a beauty app.

Who we build for first (the beachhead)

We do not launch “for everyone.” We win one underserved group loudly, then expand. The obvious, defensible beachhead is the population the incumbents serve worst: people with medium-to-deep skin tones (Fitzpatrick IV–VI) and the “forgotten middle,” who are tired of tools that misread their skin. That community is vocal, loyal when served well, and gives us the exact data that strengthens our core moat. Land there; expand outward.

POSITIONING ONE-LINER TO TEST IN ADS

“Finally, a skin scan built for *your* skin — not just the faces the other apps were trained on.”

SECTION 06

“Identify all races” — let's reframe this together

You asked for the app to “identify all races.” I want to push on this as your co-founder, because how we frame it is genuinely the most important design decision in the product — and getting it wrong is both a technical dead-end and a legal/PR landmine. Here's the honest version.

What you actually want (and it's better than what you asked for)

The goal isn't to have the AI *classify a user's race*. That's the wrong target for three reasons: (1) race is a social category, not a skin-biology one — models that predict it are brittle and offensive when wrong; (2) it invites discrimination claims and biometric-profiling scrutiny; (3) it doesn't even help the recommendation. What actually drives good skincare advice is **skin tone, undertone, and how skin behaves** — not ethnicity labels.

So we replace “identify race” with the thing that's both more useful and more defensible:

- **Measure skin tone on the Fitzpatrick scale (I–VI)** plus undertone (warm/cool/neutral) — these are legitimate skincare inputs.
- **Guarantee equal performance across all of them.** The promise is not “we can tell what you are”; it's “we work equally well no matter your skin tone,” backed by published per-band accuracy.
- **Validate per subgroup, every release.** We hold ourselves to a max accuracy gap between the best- and worst-performing skin-tone band. If the gap widens, we don't ship.

THE REFRAME, IN ONE LINE

Don't build an app that **guesses your race**. Build the one app that **works perfectly regardless of it** — and can prove it. Same ambition you had; far stronger product and brand.

On “identify all skin conditions and skin types”

Same spirit, one critical boundary. We can and should detect **cosmetic concerns** — dryness, oiliness, visible texture, the *appearance* of acne, dark spots, redness, fine lines, pores, dark circles — and classify skin **type** (dry/oily/combination/sensitive). What we must not do is **diagnose disease**: telling someone they “have rosacea” or flagging a mole as cancer crosses the line from cosmetics into regulated medicine. We describe what the skin *looks like* and recommend over-the-counter care; we never diagnose. If we see something concerning, the right move is a gentle “consider seeing a dermatologist,” not a verdict. (Full reasoning in Section 9.)

SECTION 07

The product: MVP (v0)

The temptation — yours and mine — is to build everything that beats everyone. That's how v0s die. The MVP's only job is to prove two things: **(a)** our read is good and trusted across skin tones, and **(b)** people come back. Everything else waits.

The core loop (this is the whole app at launch)

1	Scan Guided face capture with a lighting/framing check so the photo is actually usable. Front camera, 10 seconds.
2	Read AI returns a clear, friendly read: skin type, tone/undertone, and 6–8 cosmetic concern scores (hydration, texture, shine, dark spots, redness, fine lines, pores, dark circles). Plain language, no jargon, no diagnosis.
3	Recommend A simple AM/PM routine: the <i>kinds</i> of products and key ingredients to use (and avoid), with 1–3 brand-neutral product examples per step across price tiers. The 'why' is always shown.
4	Return Re-scan in ~2–4 weeks; see an honest trend; routine adjusts. A light 'did this help?' prompt feeds our outcome loop.

In scope for v0

- Guided scan + the read + a brand-neutral routine, validated across skin tones from day one.
- Progress re-scan with *honest* trend tracking.
- Consent & privacy flow built in from the first commit — not retrofitted (Section 9).
- A basic skin-coach chat to answer “why this product?” / “can I use these together?”

Deliberately OUT of scope for v0 (build later, on purpose)

- Disease diagnosis or anything cancer-adjacent (regulatory line — never in a consumer MVP).
- AR try-on / makeup overlays (Tier-1 players own this; not our wedge).
- Custom-formulated products, DNA tests, hardware devices.
- B2B/white-label SDK — a great *later* business, a distraction now.
- Training our own foundation model (Section 8 explains what we do instead).

SECTION 08

How the AI actually works (the realistic version)

You asked for an app that scans the face, has the AI read it, and “trains itself on data.” All achievable — but let’s be precise about what we build vs. buy, so you can brief the developer accurately and we don’t set fire to our runway training models we don’t need.

Three layers

Layer 1 — Capture. On-device face detection and a quality gate (lighting, focus, framing, single face). This is the cheap, high-leverage part most apps skimp on — and the reason their tracking is junk. We get it right. Standard on-device vision libraries handle this; no custom model needed.

Layer 2 — The read (the “brain”). v0 stands on existing vision models / skin-analysis APIs plus a curated rules layer for the cosmetic scores. We do NOT train a model from scratch initially — that needs data we won’t have on day one. The key engineering discipline: *evaluate any model/API we use on a balanced skin-tone test set before trusting it*. If an off-the-shelf API fails on Fitzpatrick V–VI (many do), that’s disqualifying for us specifically — because equity is the product.

Layer 3 — The recommender. Maps the read → ingredients/routine → specific products, brand-neutrally. This is where our taste and trust live, and it’s mostly a well-designed, expert-reviewed knowledge base — not exotic ML. A licensed dermatologist or cosmetic chemist advising here is worth more than a bigger model.

The data flywheel (how we eventually out-learn them)

“Trains itself on data” is real, but it’s a *flywheel we earn*, not a switch we flip. The loop: consented scans + outcomes (“did this routine help?” + follow-up scans) → we learn what works per skin-tone/skin-type cohort → recommendations improve → better results attract more of the underserved users → more (especially deep-skin-tone) data → our equity lead compounds. Critically, every loop runs on **explicit consent** with the ability to delete — which is exactly what Section 9 requires anyway.

WHAT TO ACTUALLY TELL THE DEVELOPER ABOUT “AI”

v0 = solid capture + a vetted analysis model/API (tested for skin-tone fairness) + an expert-built recommendation engine + a consented data-collection loop. NOT a from-scratch model. We graduate to fine-tuning / our own model once the flywheel has produced a balanced, labeled dataset worth training on — and we can prove it beats what we license. Sequence matters.

SECTION 09

Regulatory & privacy — do not skip this

I'm putting this in bold as your co-founder: **this section is more likely to kill the company than any competitor is.** Two issues. One governs what the app may legally say; the other governs how we may legally *collect a face* — and it's especially sharp because we're building in Illinois. Read both before a line of code ships. (Standard caveat: I'm your co-founder, not your lawyer — we retain a real one before launch. But here's what we're walking into.)

Issue A — The FDA “medical device” line

Software that helps you maintain a **healthy lifestyle / appearance** and is unrelated to diagnosing or treating disease is **not** a regulated medical device (general-wellness exemption, reaffirmed in FDA's 2026 guidance). Software that **diagnoses, treats, or detects disease** is. The entire difference is in the claims we make.

SAFE — cosmetic / wellness language	DANGER — device-triggering language
“Your skin looks dehydrated.”	“You have [a disease].”
“Visible texture / the appearance of blemishes.”	“This is acne vulgaris / rosacea / eczema.”
“Consider a gentle exfoliant; here's why.”	“This will treat / cure your condition.”
“This spot is new — consider seeing a dermatologist.”	“This mole is (likely) melanoma.”

Rule for the whole team: we describe *appearance* and recommend *over-the-counter care*. We never name a disease, never claim to treat one, and never do cancer detection. That single discipline keeps us out of FDA device territory and lets us move at startup speed. The moment a feature crosses it, it needs legal sign-off — no exceptions.

Issue B — Biometric privacy (BIPA). This one is existential.

WHY THIS IS RED, NOT YELLOW — AND WHY YOUR LOCATION MATTERS

A face scan that maps facial geometry is **biometric data**. Illinois' BIPA is the strictest such law in the U.S., it gives **private individuals the right to sue** (no regulator needed), and it does not require any actual harm. We are building in Chicago. Plaintiffs' firms watch this space full-time. Get the consent flow wrong and a single class action can end us before we raise a Series A.

What BIPA requires — build these into v0, not later:

- **Written, informed consent BEFORE capture.** Tell the user, in writing, that we collect a biometric identifier, why, and for how long — and get explicit opt-in. No pre-checked boxes.
- **A published retention & destruction policy.** State how long we keep face data and when we destroy it (and then actually destroy it on schedule).

- **Never sell or profit from biometric data.** BIPA flatly prohibits it. (Aligns with our “never sell your face” promise anyway.)
- **Reasonable security** and no disclosure without consent.

Penalties are statutory — historically \$1,000 (negligent) / \$5,000 (reckless) per person (a 2024 amendment moved this to per-person rather than per-scan, which helps, but the exposure is still enormous at scale). For scale: Facebook settled a BIPA face-tagging case for \$650M; Google for \$100M; TikTok for \$92M. We will not have their balance sheet.

How we turn this into an advantage

- **Process on-device or ephemerally** wherever feasible — if a raw face image never leaves the phone, or is deleted immediately after analysis, our exposure drops dramatically.
- **Lead with privacy in the UI.** “Your face never leaves your device / we never sell it / delete everything in one tap.” That’s a trust feature competitors can’t easily copy.
- **Get the consent language lawyer-drafted** before launch, and treat GDPR (EU) and CCPA/CPRA (California) as the next tiers once we expand. New BIPA-style laws are pending in other states — building to the strict standard now means we’re ready.

SECTION 10

Business model & pricing

Three revenue paths exist; they're not equally compatible with our trust positioning. My recommendation: lead with subscription, add commerce carefully, keep B2B2C as the big later prize.

Model	How it works	Fit with us
Freemium subscription (lead with this)	Free: one scan + basic read. Paid (~\$6–10/mo): full routine, progress tracking, coach chat, re-scans, ingredient checks.	Strong. Aligns incentives with the user, not a brand. Predictable revenue.
Commerce / affiliate (careful)	Earn a margin when users buy recommended products. Must be transparent and never distort the recommendation.	Tension. Real money, but it's the exact thing we criticize. Only with full disclosure + neutral ranking.
B2B2C / white-label (the later prize)	License our skin-tone-fair engine to brands, clinics, retailers as an SDK — the Tier-1 business, but with our equity edge.	Big upside, later. Needs a proven engine first. Likely our acquisition story.

My recommendation

Launch **freemium subscription** — it's the only model that keeps “trust” honest. Layer in **fully-disclosed commerce** once we have scale and have earned the benefit of the doubt. Treat **B2B2C licensing** as the long game and likely the exit. Resist the urge to monetize the face data itself — it's both illegal under BIPA and the fastest way to torch the brand we're building.

SECTION 11

Roadmap & the 90-day plan

Phasing keeps us honest about sequence. Equity validation and the consent flow are in Phase 1 because they are the product, not polish.

Phase	Goal	Ships
P1 · 0–3 mo Prove the read	A trustworthy, skin-tone-fair scan + read + routine, with consent built in.	Core loop (Sec 7) · balanced test set · BIPA consent flow · ~50–100 closed beta users across all 6 tones
P2 · 3–6 mo Prove retention	People come back and trust the trend. Start the outcome loop.	Honest progress tracking · coach chat · 'did it help?' loop · public launch to the beachhead
P3 · 6–12 mo Prove the moat	The data flywheel makes recs measurably better; begin our own model.	Cohort-tuned recommendations · published accuracy-by-skin-tone · fine-tuning on our consented data
P4 · 12 mo+ Scale & license	Expand beyond the beachhead; open the B2B2C path.	Broader marketing · GDPR/CCPA readiness · white-label SDK pilots

What to hand the developer on Monday

- **Build the core loop only** (Section 7): guided capture → read → brand-neutral routine → re-scan. Nothing else.
- **Wire the consent + retention/delete flow first** (Section 9-B). It is a hard launch blocker, not a v2 nicety.
- **Pick an analysis model/API, then immediately test it on a balanced skin-tone image set.** If it underperforms on Fitzpatrick V–VI, reject it — that's our whole differentiation.
- **Keep all skin language cosmetic** (Section 9-A). No disease names anywhere in the copy or model outputs.
- **Instrument the outcome loop early** (a simple 'did this help?' + scheduled re-scan), even if we don't use the data yet — it's the seed of the moat.

SECTION 12

Open questions for us (co-founder decisions)

Things I can't decide alone — let's lock these before we brief anyone:

- **Beachhead, confirmed?** Do we agree to launch *for the underserved-skin-tone community first* rather than “everyone”? (I strongly recommend yes — it's our edge.)
- **Commerce now or later?** Pure subscription at launch keeps trust clean. Are we disciplined enough to leave affiliate money on the table early?
- **On-device vs. cloud read?** On-device slashes BIPA risk but constrains model choice. This is an architecture decision with legal consequences — we should make it consciously.
- **Do we bring on a dermatologist / cosmetic chemist advisor?** I think the recommendation engine needs one for credibility and safety. Equity or cash?
- **Name.** ‘TrueTone’ is a working codename that nicely signals the skin-tone-fairness promise — but we should check trademark availability (Apple uses ‘True Tone’ for displays, which could be a conflict). Keep, or explore alternatives?
- **Budget & runway.** What can we actually spend on the first 6 months — and do we bootstrap the beta or raise a small pre-seed first?

BOTTOM LINE FROM YOUR CO-FOUNDER

The idea is good and the timing is right. The winning move isn't “every feature, every face” on day one — it's **one honest, skin-tone-fair scan that a specific underserved group falls in love with**, built on a consent flow that keeps us alive in Illinois. Nail that, and the flywheel, the premium features, and the B2B licensing all follow. Let's go build it.

Sources informing this brief include 2026 market analyses (Coherent Market Insights, Grand View, Future Market Insights), competitor materials (Haut.AI, Perfect Corp, Lovi and others), peer-reviewed work on dermatology-AI skin-tone bias (NCBI/arXiv/Wiley, 2025–2026), FDA's January 2026 general-wellness & CDS guidance, and 2025–2026 BIPA litigation reviews. Figures are directional; legal specifics require counsel before launch.